

Question 1

$$a) \quad x(t) = \beta t^2 + 5$$

$$v(t) = 2\beta t$$

$$a(t) = 2\beta \quad \checkmark \checkmark$$

$$F = ma = 2 \text{ kg} (2\beta) \\ = 4\beta \quad \checkmark$$

$$b) \quad v_0 = 0 \text{ m/s}$$

$$v_5 = 2\beta t = 2\beta(5) = 10\beta$$

$$\Delta K = 200 \text{ J} = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 \quad \checkmark$$

$$= \frac{1}{2} (2 \text{ kg}) (10\beta)^2 - 0 \quad \checkmark$$

$$= 100\beta^2$$

$$\beta = \sqrt{2} = 1.4 \quad \checkmark$$

$$c) \quad W = \Delta K \quad \checkmark$$

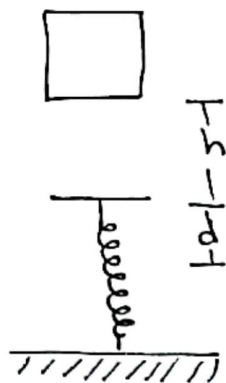
This question specifically stated that you were not to use kinematics. So the correct way to do this problem is to use the Work kinetic energy theorem,

$$F \Delta x = \Delta K$$

$$\Delta x = \frac{\Delta K}{F} = \frac{200 \text{ J}}{4(1.4) \text{ N}} = 35.4 \text{ m} \quad \checkmark$$

[8]

Question 2



Define the system:

Block
Spring
Earth

$U_g = 0, y = 0$ ✓
System is isolated. ✓

In energy problems, it is essential that you state the following:
* where the gravitational potential is zero, or where $y = 0$.
* what your system consists of and if it is isolated or not.

Conservation of Energy:

$$\Delta E = \Delta K + \Delta U_g + \Delta U_s + \Delta E_{\text{int}} = 0$$

$$0 = K_f - K_i + mg(y_0) - mg(h+d) + \frac{1}{2}kd^2 - \frac{1}{2}kx_0^2$$

$$= 0 - 0 + 0 - mg(h+d) + \frac{1}{2}kd^2 - 0$$

$$h = \frac{-mgd + \frac{1}{2}kd^2}{mg}$$

$$= \frac{-2 \text{ kg} (9.8 \text{ m/s}^2) (0.11 \text{ m}) + \frac{1}{2} (1960 \text{ N/m}) (0.11 \text{ m})^2}{2 \text{ kg} (9.8 \text{ m/s}^2)}$$

$$= 0.495 \text{ m} = 0.50 \text{ m}$$

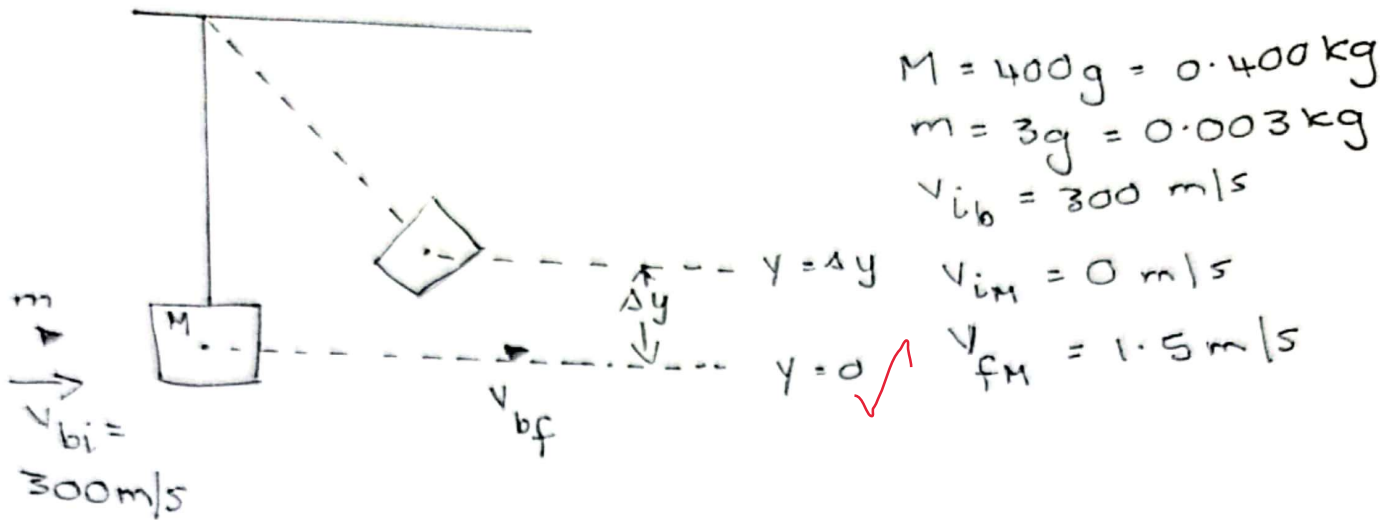
$$= 49.5 \text{ cm} \approx 50 \text{ cm}$$

The most common error in this question was that some students did not remember that as the spring compresses, the gravitational potential energy also changes. This is why it is essential to label all the significant points on the diagram, and that you realise that the final position of the block is not at the top of the uncompressed spring, but at the lowest point.

Depending on where you chose $y = 0$, the substituted values may change, but the final answer will be the same.

[6]

Question 3



a) Conservation of momentum:

$$mv_{bi} + Mv_{Mi} = Mv_{Mf} + mv_{bf} \quad \checkmark$$

$$v_{bf} = \frac{mv_{bi} + Mv_{Mi} - Mv_{Mf}}{m} \quad \checkmark$$

$$= \frac{(3 \times 10^{-3} \text{ kg})(300 \text{ m/s}) + (0.400 \text{ kg})(0 - 1.5 \text{ m/s})}{3 \times 10^{-3} \text{ kg}}$$

$$= 100 \text{ m/s} \quad \checkmark$$

b) Conservation of energy

System: Isolated: Block + Earth. $\checkmark$

$$\Delta E = \Delta K + \Delta U + \Delta E_{\text{int}} = 0 \quad \checkmark$$

$$K_f - K_i + U_{gf} - U_{gi} = 0$$

$$0 - \frac{1}{2}mv_i^2 + mgy_f - mgy_i = 0 \quad \checkmark$$

$$y_f = \frac{\frac{1}{2}mv_i^2}{mg} = \frac{\frac{1}{2}v_i^2}{g} = 0.115 \text{ m} \quad \checkmark$$

Question 3 cont.

$$\begin{aligned} (c) \quad W &= \Delta K \\ &= \frac{1}{2} m_b v_f^2 - \frac{1}{2} m_b v_i^2 \quad \checkmark \\ &= \frac{1}{2} (0.003 \text{ kg}) ((100 \text{ m/s})^2 - (300 \text{ m/s})^2) \\ &= -120 \text{ J} \quad \text{work done on bullet.} \end{aligned}$$

Work done by bullet is 120 J. $\checkmark$

The work kinetic energy theorem gives you the work done on the bullet, and gives you a negative value for the work done. The work done by the bullet is just the negative of the work done on the bullet - giving you a positive value for the work done.

(d) System: bullet + block.
Initial is just before collision.
Final is just after collision.

$$\Delta E = \Delta K + \Delta U + \Delta E_{\text{int}} = 0$$

$$\Delta K_b + \Delta K_M + \Delta U_b + \Delta U_M + \Delta E_{\text{int}} = 0 \quad \checkmark$$

$$\frac{1}{2} m_b (v_f^2 - v_i^2) + \frac{1}{2} M (v_{fM}^2 - v_{iM}^2) = -\Delta E_{\text{int}} \quad \checkmark$$

$$-120 \text{ J} + \frac{1}{2} (0.400 \text{ kg}) (1.5 \text{ m/s}^2 - 0^2) = -\Delta E_{\text{int}}$$

$$\therefore \Delta E_{\text{int}} = +120 \text{ J} - 0.45 \text{ J} = 119.55 \text{ J} \quad \checkmark$$

In the last part of the question, we need to look at the entire system, Bullet, and block. In part c of the question we already calculated the change in the kinetic energy of the bullet, but just after the collision, there is also a change in the kinetic energy of the block which must be taken into account. For the final answer to be correct - even if you rounded off to 120 J, you need to have shown that the change in the energy of the block was taken into account.

[12]

Question 4

- a) Forces: Weight of cylinder ✓
External force pulling string. resulting in Tension ✓



$$\Sigma F_y = ma_y = 0 = T - mg \quad \checkmark$$
$$T = mg$$

For part a - the net force = 0 N, and you need to state Newton's 2nd law and then apply it to the situation.

b) $\tau = \tau_1 + \tau_2 = I\alpha \quad \checkmark$

$$\alpha = \frac{1}{I} [R_1(mg) + R_2 T] \quad \checkmark \quad \text{But } R_1 = 0$$

$$= \frac{1}{\frac{1}{2}MR_2^2} [R_2(mg)] \quad \checkmark \quad R_2 = R = \text{radius.}$$

$$\alpha = \frac{2g}{R} \quad \checkmark$$

Here again we need to state Newton's 2nd law for rotation and then apply it to the situation. It is important to realise that although the centre of mass of the cylinder is not accelerating, there is angular acceleration. Using the tension obtained in (a) the acceleration is independent of the mass of the cylinder.

c) $a = \alpha R \quad \checkmark = \left(\frac{2g}{R}\right) R \quad [\text{sub from (b)} \quad \checkmark]$

$$a = 2g \quad \checkmark$$

Even though there is no net acceleration of the cylinder, the string that is unwound from the cylinder accelerates upwards, otherwise the cylinder would fall to the ground. This acceleration is related to the angular acceleration.

[9]

Question 5

a) $A = 5.00 \mu\text{m}$

For oscillator: $y = A \cos(\omega t + \phi)$

$$a = -\omega^2 A \cos(\omega t + \phi)$$

$$a_{\max} = \omega^2 A = g \quad \checkmark$$

$$\omega = \sqrt{\frac{g}{A}}$$

$$\omega = 2\pi f = \sqrt{\frac{g}{A}} \quad \checkmark$$

$$f = \frac{1}{2\pi} \sqrt{\frac{g}{A}} = \frac{1}{2\pi} \sqrt{\frac{9.8 \text{ m/s}^2}{5 \times 10^{-6} \text{ m}}}$$

$$= 223 \text{ Hz} \quad \checkmark$$

You need to state that this is the maximum acceleration of the diaphragm and relate it to the amplitude of the oscillation.

b) $\omega^2 A = a_{\max}$

$$(2\pi f)^2 A = a_{\max} \quad \checkmark$$

If f is doubled $a_{\max}$ will be 4 times higher ✓

A specific number is needed here - if the frequency is doubled the acceleration will be 4 times larger.
A reason is also needed to substantiate your answer.

[5]

Question 6

$$y_1 = 2.00 \sin(20.0x - 32.0t)$$

$$y_2 = 2.00 \sin(25.0x - 40.0t)$$

$$\begin{aligned} a) \Delta\phi &= -(20.0x - 32.0t) + (25.0x - 40.0t) \\ &= (5.00x - 8.00t) \end{aligned}$$

$$\text{At } x = 5.00 \text{ cm and } t = 2.00 \text{ s}$$

$$|\Delta\phi| = 9 \text{ radians} \times \frac{360^\circ}{2\pi \text{ rad}} = 515.6^\circ$$

$$\Delta\phi = 158^\circ$$

$$b) \Delta\phi = (5.00x - 8.00t)$$

$$\text{When } \Delta\phi = (\pm\pi) \text{ or } (\pm\pi + n2\pi) \quad y_1 + y_2 = 0.$$

$$\text{For } t = 2.00 \text{ s:}$$

$$\Delta\phi = (5.00x - 16) = (\pm\pi + 2n\pi)$$

$$\begin{aligned} \text{For } n = 0; \text{ and } +\pi \\ x &= \approx 9.57 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{For } n = 1: \\ x &= 25.4 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{For } n = 0, \text{ and } -\pi \\ x &= 2.57 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{For } n = 1 \text{ (}\Delta\phi \text{ neg.)} \\ x &= 1.32 \text{ cm} \end{aligned}$$

$$\begin{aligned} n = 2 \\ x &= 0.058 \text{ cm} \end{aligned}$$

For $+\Delta\phi$
values increase

For $-\Delta\phi$
values decrease
initially.

The critical part of this question is to realise that the phase difference needs to be π , but the sine function repeats after 2π radians. You cannot say that it is just $n\pi$, because for all even values of n , the waves are in phase.

Question 7

Energy stored in the bullet as kinetic Energy

$$K = \frac{1}{2} m v^2$$

$$\Delta K = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$
$$= -\frac{1}{2} m v_i^2 \quad \checkmark$$

To melt bullet:

Heat bullet to melting point:

$$Q_1 = m c \Delta T \quad \checkmark$$

~~To~~ Melt

$$Q_2 = m L_F \quad \checkmark$$

If all KE is converted to heat then:

$$Q_1 + Q_2 + \Delta K = 0 \quad \checkmark$$

$$Q_1 + Q_2 = -\Delta K$$

$$m c \Delta T + m L_F = \frac{1}{2} m v_i^2$$

$$2 [c \Delta T + L_F] = v_i^2 \quad \checkmark$$

$$\left[2 (128 \text{ J/kg} \cdot ^\circ\text{C}) (327^\circ\text{C} - 25^\circ\text{C}) + 2 \cdot 32 \times 10^4 \text{ J/kg} \right]^{\frac{1}{2}} = v_i \quad \checkmark$$

$$v_i = 352 \text{ m/s} \quad \checkmark$$

This is plausible, but energy is needed to break the wall in order to embed. $\checkmark$

Your final answer needs to be supported by some or other argument for or against the plausibility of the myth